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Object-Centric Process Mining

Some processes do not have one natural case identifier.

The flattening problem

One order may contain

- ▶ Several items
- ▶ Several deliveries
- ▶ Several invoices
- ▶ One or more payments

Forced order case

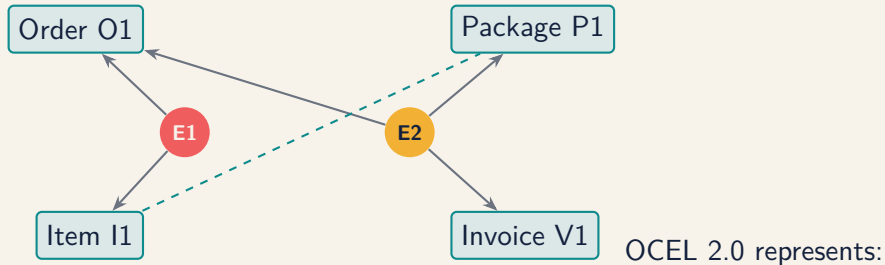
Item-level events may be duplicated across the flattened trace.

Forced item case

Shared invoice and delivery events may be duplicated instead.

This creates **convergence** and **divergence**: misleading frequencies and apparent behavioral relations.

Object-centric event data



- ▶ Events linked to one or more typed objects
- ▶ Object-to-object relationships
- ▶ Qualified relationships and changing object attributes

Case-centric or object-centric?

Situation	Case-centric	Object-centric
One stable business instance	Strong fit	Often unnecessary
Several interacting entities	Distorting	Strong fit
Established KPI by case	Direct	Requires aggregation choices
Familiar tooling	Mature	Emerging
Relationship questions	Limited	Native

Key idea Object-centric analysis does not eliminate modeling choices. It makes relationships explicit so those choices are visible.

Object-centric features without premature flattening

Per object type

- ▶ Counts and lifecycle states
- ▶ Age and elapsed-time summaries
- ▶ Completed and open related objects
- ▶ Relationship cardinalities

At prediction time

- ▶ Freeze the object graph at the cutoff
- ▶ Aggregate only visible objects
- ▶ Preserve identifiers for grouped splits
- ▶ Test sensitivity to aggregation rules

Key idea Flatten late and explicitly. The target object and prediction time should determine which relationships become features.

Inspect an OCEL with PM4Py

```
import pm4py

ocel = pm4py.read_ocel2_json("order_management.jsonocel")

print(ocel.events.head())
print(ocel.objects.head())
print(ocel.relations.head())

ocdfg = pm4py.discover_ocdfg(ocel)
pm4py.save_vis_ocdfg(ocdfg, "object_centric.svg")
```

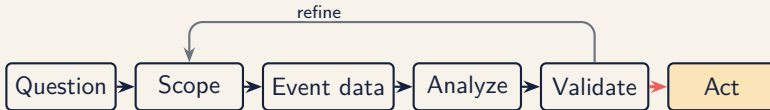
Lab comparison: flatten the same data by order and by item. Which frequencies or paths become misleading?

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Projects and Communication

A process-mining project succeeds when evidence changes a decision.

The project lifecycle



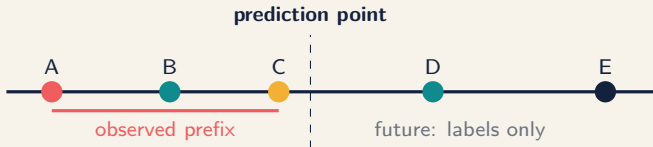
The lifecycle is iterative. Domain validation may change the event mapping, case notion, model, metric, or even the original question.

Predictive process monitoring tasks

Question	ML formulation	Example target	Useful baseline
What happens next?	Classification	Next activity	Most frequent transition
How long remains?	Regression / survival	Remaining time	Median by current state
How will it end?	Classification	Late / rejected / compliant	Base-rate model
Is this unusual?	Anomaly detection	Deviant prefix	Frequency / conformance score

Key idea The prediction must correspond to an operational decision and a time when someone can still act.

Prefixes, labels, and prediction points



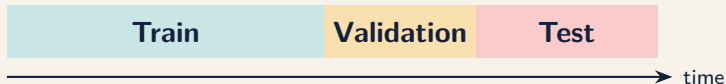
Define

- ▶ Prefix rule and cutoff
- ▶ Prediction frequency
- ▶ Label horizon

Avoid

- ▶ Features computed after cutoff
- ▶ Labels unavailable in operation
- ▶ Duplicated prefixes across splits

Split by time, group by case



- ▶ Assign every case and all its prefixes to exactly one split.
- ▶ Fit encoders, imputers, scalers, and vocabularies on training data only.
- ▶ Prefer temporal evaluation when the deployed model predicts future cases.
- ▶ Keep related customers, orders, or objects together when dependence exists.
- ▶ Reserve the test period for one final estimate after tuning.

Common leakage paths

Future leakage

- ▶ Final case attributes in a prefix
- ▶ Total event count or final duration
- ▶ Post-outcome resource or status

Split leakage

- ▶ Prefixes from one case in several splits
- ▶ Random event-level splitting

Preprocessing leakage

- ▶ Global normalization or imputation
- ▶ Variants built from all periods
- ▶ Discovery model fitted on test cases

Operational leakage

- ▶ Feature not available at scoring time
- ▶ Label definition changes after deployment

Evaluate beyond predictive accuracy

Discrimination

AUC, PR-AUC,
MAE, concordance

Calibration

Reliability, Brier
score, interval
coverage

Timeliness

Earliness, stability
across prefixes

Utility

Cost, workload,
prevented delay,
decision value

Also monitor:

- ▶ Performance by cohort and process state
- ▶ Feature, label, and process drift
- ▶ Alert volume, false-alarm burden, and intervention capacity

Integration patterns: process mining + ML

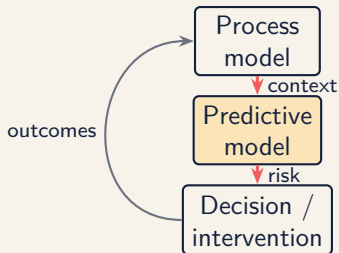
Process mining informs ML

Variants, model state, alignments, conformance, rework, and handovers become features or cohorts.

ML informs process analysis

Predictions identify risky prefixes and reveal where errors concentrate in the process.

Key idea Version and monitor the event mapping, process model, encoding, and predictive model as one analytical system.



An evidence chain for every finding

1. **Question:** what decision is at stake?
2. **Observation:** what pattern is measured?
3. **Interpretation:** what mechanism may explain it?
4. **Validation:** what did domain experts or other data confirm?
5. **Action:** what change should be tested?
6. **Metric:** how will we know whether it worked?

Key idea Separate observation from interpretation. It makes the analysis easier to challenge, improve, and trust.

Communicate one finding well

Lead with

- ▶ Operational consequence
- ▶ Magnitude and affected population
- ▶ Concise visual evidence
- ▶ Recommended next action

Always include

- ▶ Data scope
- ▶ Key assumptions
- ▶ Uncertainty
- ▶ Important alternatives

Headline pattern

What happens + **where** + **impact**: “Clarification loops add 4.1 days to partner-channel custom orders.”

Reproducibility checklist

- ▶ Versioned extraction and transformation code
- ▶ Data dictionary and event semantics
- ▶ Fixed parameters and package versions
- ▶ Deterministic figures and tables
- ▶ Documented exclusions and filters
- ▶ Stored model artifacts
- ▶ Decision and validation log
- ▶ Privacy-aware sharing strategy

Discuss Could another analyst reproduce your central finding without asking what you clicked?

Research opportunities

Richer event data

- ▶ Multimodal event logs
- ▶ Automated event abstraction
- ▶ Object-centric, streaming, and uncertain events

Richer analysis

- ▶ Multimodal representation learning
- ▶ Causal process mining
- ▶ Prescriptive monitoring
- ▶ Privacy-preserving analytics

Multimodal event logs

Align structured events with payload such as free text, documents, images, audio, and sensor streams while preserving time, provenance, and access rules.

Key idea Each modality must add valid, timely, and governable process evidence.